

6. Reduce each of the following matrices into the indicated form:

(i)
$$\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{bmatrix}$$
 reduced echelon form

Sol.
$$\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{bmatrix} \xrightarrow{R} \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 3 & -4 & 4 \\ 0 & 7 & -7 & 6 \end{bmatrix}$$
 by $R_2 - 2R_1$
and $R_3 - 3R_1$

$$\xrightarrow{R} \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 3 & -4 & 4 \\ 0 & 1 & 1 & -2 \end{bmatrix} \text{ by } R_3 - 2R_2$$

$$\xrightarrow{R} \begin{bmatrix} 1 & 0 & 5 & -5 \\ 0 & 0 & -7 & 10 \\ 0 & 1 & 1 & -2 \end{bmatrix} \text{ by } R_1 + 2R_3 \text{ and } R_2 - 3R_3$$

$$\xrightarrow{R} \begin{bmatrix} 1 & 0 & 5 & -5 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & -7 & 10 \end{bmatrix} \text{ by } R_{23}$$

$$\underset{\sim}{R} \begin{bmatrix} 1 & 0 & 5 & -5 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & -\frac{10}{7} \end{bmatrix} \quad \text{by } -\frac{1}{7} R_3$$

$$\underset{\sim}{R} \begin{bmatrix} 1 & 0 & 0 & \frac{15}{7} \\ 0 & 1 & 0 & -\frac{4}{7} \\ 0 & 0 & 1 & -\frac{10}{7} \end{bmatrix} \quad \begin{array}{l} \text{by } R_1 - 5R_3 \\ \text{and } R_2 - R_3 \end{array}$$

which is the required reduced echelon form of the given matrix.

$$6.(ii) \quad \begin{bmatrix} 0 & 1 & 3 & -2 \\ 2 & 1 & -4 & 3 \\ 2 & 3 & 2 & -1 \end{bmatrix} \quad \text{reduced echelon form}$$

$$\text{Sol.} \quad \begin{bmatrix} 0 & 1 & 3 & -2 \\ 2 & 1 & -4 & 3 \\ 2 & 3 & 2 & -1 \end{bmatrix} \underset{\sim}{R} \begin{bmatrix} 0 & 1 & 3 & -2 \\ 2 & 1 & -4 & 3 \\ 0 & 2 & 6 & -4 \end{bmatrix} \quad \text{by } R_3 - R_2$$

$$\underset{\sim}{R} \begin{bmatrix} 0 & 1 & 3 & -2 \\ 2 & 0 & -7 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{by } R_2 - R_1 \\ \text{and } R_3 - 2R_1 \end{array}$$

$$\underset{\sim}{R} \begin{bmatrix} 2 & 0 & -7 & 5 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{by } R_{12}$$

$$\underset{\sim}{R} \begin{bmatrix} 1 & 0 & -\frac{7}{2} & \frac{5}{2} \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{by } \frac{1}{2} R_1$$

is the required reduced echelon form of the given matrix.

$$6.(iii) \quad \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 2 & 4 & 1 & -2 & 3 \\ 3 & 6 & 2 & -6 & 3 \end{bmatrix} \quad \text{echelon form}$$

$$\text{Sol.} \quad \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 2 & 4 & 1 & -2 & 3 \\ 3 & 6 & 2 & -6 & 3 \end{bmatrix} \underset{\sim}{R} \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 3 & -6 & 1 \\ 0 & 0 & 5 & -12 & 0 \end{bmatrix} \quad \begin{array}{l} \text{by } R_2 - 2R_1 \\ \text{and } R_3 - 3R_1 \end{array}$$

$$\tilde{R} \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 1 & -2 & \frac{1}{3} \\ 0 & 0 & 1 & -\frac{12}{5} & 0 \end{bmatrix} \text{ by } \frac{1}{3} R_2 \text{ and } \frac{1}{5} R_3$$

$$\tilde{R} \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 1 & -2 & \frac{1}{3} \\ 0 & 0 & 0 & -\frac{2}{5} & -\frac{1}{3} \end{bmatrix} \text{ by } R_3 - R_2$$

$$\tilde{R} \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 1 & -2 & \frac{1}{3} \\ 0 & 0 & 0 & 1 & \frac{5}{6} \end{bmatrix} \text{ by } -\frac{5}{2} R_3$$

$$\tilde{R} \begin{bmatrix} 1 & 2 & -1 & 0 & -\frac{2}{3} \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & \frac{5}{6} \end{bmatrix} \text{ by } R_2 + 2R_3 \text{ and } R_1 - 2R_3$$

is the required echelon form of the given matrix.

$$6.(iv) \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 2 & -5 & 3 & 1 \\ 4 & 1 & 1 & 5 \end{bmatrix} \text{ reduced echelon form.}$$

$$\text{Sol.} \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 2 & -5 & 3 & 1 \\ 4 & 1 & 1 & 5 \end{bmatrix} \tilde{R} \begin{bmatrix} 1 & 4 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & -11 & 5 & -3 \\ 0 & -11 & 5 & -3 \end{bmatrix} \text{ by } R_3 - 2R_1 \text{ and } R_4 - 4R_1$$

$$\tilde{R} \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_3 + R_2 \text{ and } R_4 + R_2$$

$$\tilde{R} \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 1 & -\frac{5}{11} & \frac{3}{11} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \text{ by } \frac{1}{11} R_2$$

$$\tilde{R} \begin{bmatrix} 1 & 0 & \frac{4}{11} & \frac{13}{11} \\ 0 & 1 & -\frac{5}{11} & \frac{3}{11} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \text{ by } R_1 - 3R_2$$

is the required reduced echelon form of the given matrix